

THE STORY OF MEASURING TIME

THE ACCURATE MEASUREMENT OF TIME IS VITAL TO MANY ASPECTS OF THE MODERN WORLD

The modern conception of time of a standardized quantity is shared across the world. It combines knowledge of astronomical calendars and clocks based on the apparent motions of stars and planets with recent technologies for measuring and recording relatively short intervals of time.

Humans have probably been aware of the passage of time from the dawn of consciousness, but a proper understanding of the seasons and changing length of days throughout the year only became important with the beginning of settled agriculture around 8000 BCE. Prehistoric monuments from around the world, including Stonehenge in Britain, show a clear ability to track seasons from the rising and setting of the Sun.

The need to measure smaller time intervals arose only with the advanced civilizations of ancient Mesopotamia around 2000 BCE, probably driven by religious, ritual, and administrative requirements. Sundials were used to roughly track the time of day, while shorter time intervals were

measured by tracking the dripping of water, or later the flow of sand, through a narrow aperture.

CLOCK TIME

The earliest weight-driven mechanical clocks probably originated in Europe early in the 2nd millennium CE. A single clock on a public building such as a church sufficed for an entire community. Mechanical clocks became portable with the introduction of the spring drive around 1500, and their accuracy was greatly improved in the late 17th century. The Industrial Revolution, bringing with it faster travel and telegraph communication, eventually forced a standardization of timekeeping across widespread areas.

zodiac ring shows the constellations

main hand indicates local solar time

the Sun moves through the zodiac constellations over the course of a year



Astronomical clock, Prague, Old Town Square, Czech Republic

“...TRUE AND MATHEMATICAL TIME... FLOWS UNIFORMLY AND... IS CALLED DURATION.”

Isaac Newton, from *Principia*, 1687

2000 BCE

First calendars

Ancient Babylonians develop the earliest known calendars. The year is split into 12 months based on the lunar cycle and an extra month added to bring the lunar and solar cycles back into line. Other civilizations develop similar calendars.



Mayan calendar

520 CE

Time candles

The first reference to time candles—slow-burning wax candles or sticks of incense, which can roughly reveal the time even at night—is made in a Chinese poem.

800 CE

Hourglasses

The first definitive references to this sand-based equivalent of the water clock date from the 14th century, but the sandglass was probably invented there, in the early 9th century.



Greek clepsydra

1600 BCE

Water clocks

Although probably developed in Mesopotamia, water clocks (clepsydra) become popular in Greece and Rome. Typically, a graduated marker is used to track the level of water in a container with a small hole on the base.

1500 BCE

Early sundials

Developed in both Babylon and Egypt, the first sundials track time through the shadow cast by an upright rod called a gnomon.

Ancient Egyptian sundial



1088

Su Song's clock tower

Chinese scholar Su Song builds a water clock that uses a complex series of gears to keep track of astronomical cycles, anticipating advances in European clockmaking technology.



Clock tower